

	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE		
	Winter Examination – 2022		
	Course: B. Tech.	Branch :	Semester : III
	Subject Code & Name: BTFT303 - Process Plant Material & Energy Balance		
	Max Marks: 60	Date: 13.03.2023	Duration: 3 Hr.
	Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.		
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	What is limiting and excess reactant? Explain with Example.		6
B)	Define Deston’s Law of Partial pressure and explain it.		6
C)	Difference between Wet Bulb and Dry Bulb		6
Q.2	Solve Any Two of the following.		12
A)	Name any 4 separation and explain any 1		6
B)	Define Drying and Crystallization operation and explain crystallization.		6
C)	Distinguish between Recycle and bypass steam		6
Q. 3	Solve Any Two of the following.		12
A)	Enlist the unit operations and explain any 1		6
B)	Distinguish between sensible and latent heat with an example		6
C)	Explain various thermodynamic process like constant volume, constant pressure isothermal and adiabatic process		6
Q.4	Solve Any Two of the following.		12
A)	Define the following (Molality, Molarity, Molal Friction and Volume percentage)		6
B)	Write short note on (a. Steady state, b. Unsteady state)		6
C)	Write down in detail. A mixture consisting of 15 mole % and 85 % inerts is feed to an absorption tower where it is contacted with liquid solvent B which absorbs A. The mole ratio at solvent to gas entering tower is 2:1 , The gas leaving the absorber contains 2.5% A 1.5% B rest inerts on mole basis. Calculate 1. The 1 % recovery of Solute A and 2.		6

	The fraction of solvent evaporates and gets added in gas leaving the tower.		
Q. 5	Solve Any Two of the following.		12
A)	Define the following:- 1. Relative humidity, 2. Absolute Humidity, 3. Gauge Pressure, 4. Stoichiometry		6
B)	Describe Orset Apparatus		6
C)	10,000 Kg/h of solution containing 20% methanol is continuously fed to distillation column. Distillation is found to contain 98 % methanol and waste solution from the column carries 1 % methanol. All % are by weight. Calculate: 1. The mass flow rates of Distillate and bottom product. 2. The percentage loss of Methyl alcohol		6
	*** End ***		

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